Competition and Market Definition in Local Markets

Competition Clinic

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Disclaimer: the views expressed are those of the author and cannot be regarded as stating an official position of the University of Vienna.

Outline

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  - Economics of local markets
  - Catchment areas
- Implementation
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  - Data requirements
  - Competitive assessment
- Case studies
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  - Sainsbury's–Asda (UK)
  - EDEKA/tegut (Germany, current)

Concepts

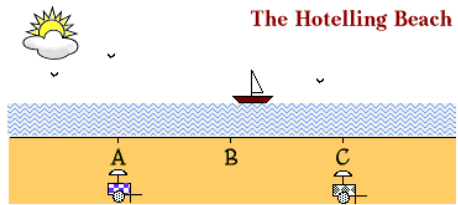
Definition: antitrust market

- Competition economists are usually interested in *antitrust markets* with two dimensions
  - Product market
  - Geographic market
- The *SSNIP test*[*] finds an antitrust market if a hypothetical price increase of 5–10% is profitable for a given set of products and area
- Conceptually, start from the narrowest market; if a SSNIP is not profitable, enlarge the set of products and the area until it is
- In practice, case law and other tools are used more often than the SSNIP test

[*] *Small but significant non-transitory increase in price* (EC Market Definition Notice 2024, para. 29).

Definition: local market

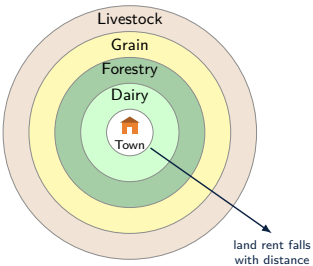
- A *local market* is a geographical area where a SSNIP starting from a minimal area is "quickly" profitable
- Space matters, i.e., actually affects exchange
- Models of spatial competition: von Thünen (1826), Hotelling (1929), Salop (1979)



Hotelling's (1929) linear city: the beach.

Models of spatial competition

von Thünen (1826)



Isolated state: land use forms rings around the town, because transport cost rises with distance

Hotelling (1929)



Linear city: consumers buy from the nearer shop; analysis of indifferent consumer

Salop (1979)



Circular city: each shop competes only with its two neighbours

Revised Market Definition Notice EC 2024 p. 6

- Product market

  *The relevant product market comprises all those **products** that customers regard as interchangeable or **substitutable** to the product(s) of the undertaking(s) involved, based on the products' characteristics, their prices and their intended use, taking into consideration the conditions of competition and the structure of supply and demand on the market.*

- Geographic market

  *The relevant geographic market comprises the geographic area in which the undertakings involved supply or demand relevant products, in which the **conditions of competition** are sufficiently **homogeneous for the effects** of the conduct or concentration under investigation **to be assessed**, and which can be distinguished from other geographic areas, in particular because conditions of competition are appreciably different in those areas.*[*]

[*]Source: European Commission (2024), Revised Market Definition Notice, C/2024/1645, p. 6.

Why engage in local market analysis?

- Many markets are local markets
  - Raw materials: raw wood, cement
  - Daily needs: gas stations, groceries, ATMs, (lawyers, ...)
- Many mergers in local markets
- Firms and authorities need precise economic analysis of local competition
- Top-down (e.g., national) analysis is too imprecise: risk of false negatives
- Competition varies across locations, which lends itself to location-specific analysis and remedies

The economics of local markets

- Demand
  - Space differentiates products through differences in transport cost
  - Transport cost: gasoline, opportunity cost of time, taste for travelling
  - May exclude consumers with a low willingness to pay from the market
  - Closer products have a transport-cost advantage
- Supply
  - Differentiation can lead to equilibrium price above marginal cost
  - Firms have an incentive to enter a profitable oligopoly
  - But available space, regulation or minimum efficient scale often limit entry
  - Some products can only be transported to a limited extent
- Main analytical tool for local market definition: the *catchment area*

Definition: catchment area

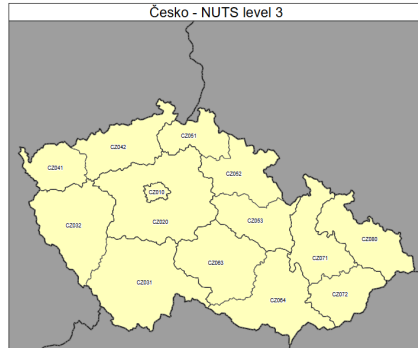
- A *catchment area* (CA) is a polygon on the map that includes all points "relevant" to a given starting point
- In economics, relevance is based on the exchange of goods and services
- Starting point
  - Supplier-centric approach
  - Customer-centric approach
- Usually, a simple rule is used to draw catchment areas around starting points
- Clustering approaches also exist, but are less common in practice
- Caveat: in theory, CAs depend on prices (at very low prices consumers travel further)

Example rule: regions

- Administrative boundaries
- Similar and often more easily available: NUTS regions*

Examples of administrative and non-administrative units being designated as NUTS regions are:

NUTS level	Germany	Spain	Czechia	Italy
NUTS 1	Bundesländer	*non-administrative aggregations*	Území	*non-administrative aggregations*
NUTS 2	Regierungsbezirke *(nonadministrative aggregations)*	Communidades autónomas	*Regiony soudržnosti (nonadministrative aggregations)*	Regioni
NUTS 3	Kreise und Kreisfreie Städte	Provincias	Kraje	Province



*Source: Eurostat (2020), *Statistical regions in the European Union and partner countries*.

Example rule: distance as the crow flies

- Consumers do not stop buying at administrative boundaries
- The outer edge of the polygon marks an *isodistance*
- Starting point plus chosen radius



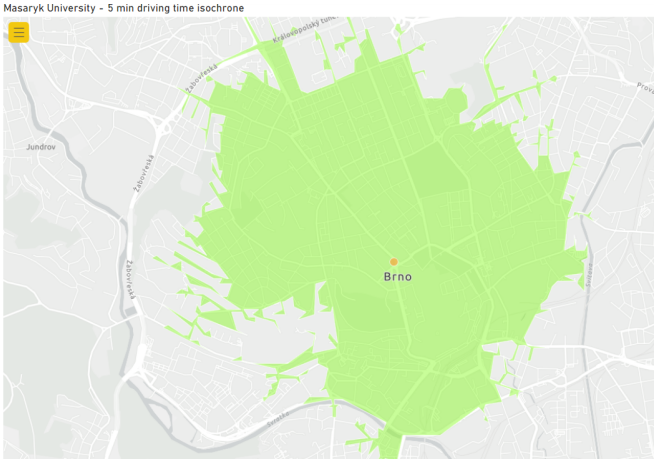
Example rule: distance based on surroundings

- Distance takes streets, rivers, mountains, etc. into account
- A reasonable approximation of driving time may be possible
- Still, CAs based on actual driving time are more precise (e.g., for gasoline)

Example rules: distance based on driving time

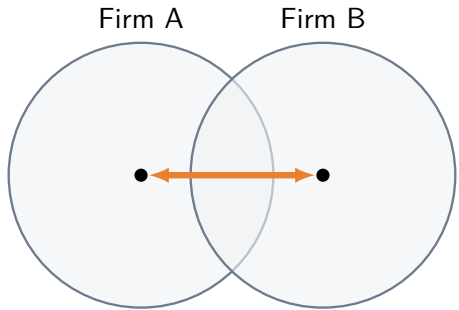
- Exact driving time depends on speed limits and congestion
- The outer edge of the polygon marks an *isochrone*
- Consider whether other transport options are practical
  - Walking
  - Ship
  - Train
  - Plane

Example isochrone



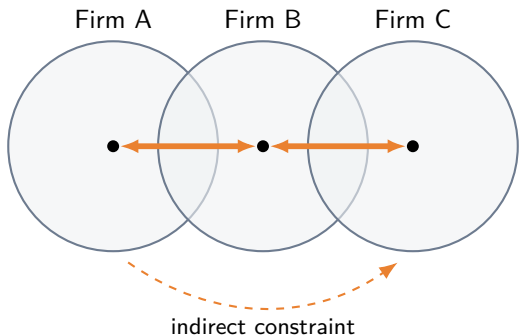
Market definition with catchment areas

- Supplier-centric approach: firms are in the same market if and only if their CAs overlap
- Supplier-centric is the main framework if firms do not price by customer location



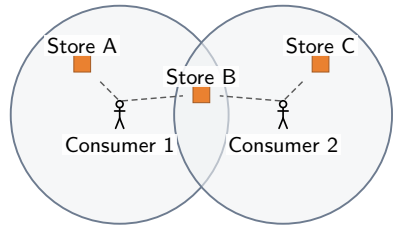
Market definition with catchment areas

- Chain of substitution: a firm constrains its neighbour, which in turn constrains the next firm



Customer-centric approach

- Catchment areas around consumers' homes: the stores inside form the choice set
- Stores compete if one consumer reaches both: A–B and B–C, not A–C
- Better reflects consumer choices and thus competition (home–store)
- In B2C markets the analysis often becomes intractable
- In B2B markets with few buyers and many sellers it is the opposite: the supplier-centric approach is intractable



Search theory and the customer-centric approach

- Consumers know only some stores and prices; finding out more costs time and travel (Stigler 1961)
- The catchment area is the consumer's *consideration set*: only stores within reach are compared
- Even small search costs can sustain monopoly prices, however many stores exist (Diamond 1971)
- Prices fall with the share of *informed* consumers, not necessarily with the number of stores (Varian 1980; Stahl 1989)
  - Commuters compare petrol prices along their route (Pennerstorfer et al. 2020)
  - Entry lowers prices most for informed consumers (Fischer, Martin and Schmidt-Dengler 2026)
- *Ordered search*: the store consumers check first gets more customers and, in theory, charges less (Armstrong, Vickers and Zhou 2009)

Implementation

Calculation of local market shares

- Number of competitors: count stores by firm in the CA
- Missing data? Be creative: e.g., predict turnover from store size, buildings or sociodemographics
- What weighting best captures competition?
  - Equal weights (count of stores)
  - Turnover or volumes
  - Capacity or floor area, e.g., quantity-area for raw wood
  - Brand, size and distance, e.g., for supermarkets
  - Demand diversion from customer surveys (CMA in Sainsbury's–Asda)
- Consider stores across the border and online retailers
- Good practice: robustness checks on radius, weighting, etc.

Special case: cities

- For some products, many CAs overlap, especially in cities
- Competition authorities have therefore treated a city (and possibly its suburbs) as a single market
- Idea: consumers move through the city every week, so they may also buy far from home (e.g., near the office)
- Clustering largely overlapping CAs can also make sense outside cities, if it simplifies the analysis without changing the conclusion

Special case: other important routes

- In some markets (e.g., gas stations), many consumers travel along fixed routes (like highways)
- Market shares and deltas along a sequence of competitors can be higher than within a CA
  - Only one firm in the CA: the CA shows no delta
  - Several firms in the CA, but some are hard to reach (e.g., other side of the highway): the CA underestimates the parties' share
- Analysing stores in the order consumers become aware of them is more realistic (compare reading Google search results)

What data and software do you need?

- Geodata: store coordinates, regional shapefiles, road network with speed limits
- Store characteristics: brand, product offering, capacity, quantities, turnover
- Customer surveys or industry data on a reasonable radius (km or minutes)
- Optional: population density, supply data, satellite images of natural resources
- Software: ArcGIS, GeoPandas, Power BI, Stata, R



Competitive Assessment

- Unilateral effects
  - Overlap analysis: market share or HHI (level and delta)
  - Chain of substitution: indirect constraints differ across pairs of firms (non-transitive)
  - Critical areas: thresholds such as market share above **40%** and delta above **20 pp**; remedy: divest stores in critical areas
  - Analysis of important routes
- Coordinated effects
  - Price ranking and price cycles
  - Timing of price changes (price leadership, reactions)
  - The same set of competitors across many CAs
- Implicit assumption so far: little differentiation other than space and uniform transport cost (a customer survey allows a better analysis if, e.g., product offering also matters)

Case Studies

Case study: EG–OMV (cleared with remedies)

- Case B8-77/21 before the German Bundeskartellamt (BKartA)
- Market: retail sale of gasoline (gas stations)
- Local market analysis identified several *critical areas*
- Remedy offered by the merging parties: divestment of **48** stations in critical areas

EG–OMV: radius

- Customer data from loyalty cards (also from non-merging competitors)
- Driving time from each station to the population centre of each customer zip code
- Cumulative turnover share by driving time, averaged over all stations:

	75%	80%	85%
Driving time (minutes)	19	27	42

Share of turnover/transactions reached within the driving time. Source: Bundeskartellamt, B8-77/21, p. 35.

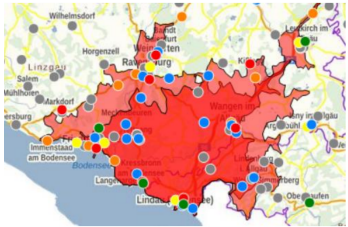
- Chosen uniform radius: 20 and 30 minutes driving time; stations across the border were not considered

EG–OMV: robustness check CA 20, 30 min

(114) Marktraum/Tankstellencluster am Bodensee mit folgenden OMV-Tankstellen:

- 88079 Kressbronn, Lindauer Str. 12
- 88131 Lindau, Kemptener Str. 14-16

(115) Einzugsbereiche des Tankstellenclusters am Bodensee, 30 Minuten Fahrzeit:



	Dieselkraftstoff in m³		Ottokraftstoff in m³	
Zieltankstelle	20 Min.	30 Min.	20 Min.	30 Min.
Kressbronn	44.000	105.000	35.000	83.000
Lindau	26.000	92.000	16.500	67.000

(301) Danach ergibt sich folgende Anteilsverteilung für das Jahr 2020:

Marktraum um Zieltankstelle	EG	OMV	EG+OMV	Aral/BP	Shell	3-er Olig.	Total	JET	ENI	Tessol (AVIA)	Sons-tige
Dieselkraftstoff 2020											
20 Minuten Fahrzeit (Anteile in %)											
Kressbronn	[5-10]	[5-10]	**[10-15]**	[30-35]	[5-10]	**[50-60]**	[0-5]	[5-10]	[5-10]	[5-10]	[15-20]
Lindau	[10-15]	[5-10]	**[20-25]**	[40-50]	[15-20]	**[80-90]**	[0-5]	[5-10]	[0-5]	[10-15]	[0-5]
30 Minuten Fahrzeit (Anteile in %)											
Kressbronn	[5-10]	[0-5]	**[10-15]**	[30-35]	[15-20]	**[50-60]**	[0-5]	[5-10]	[0-5]	[0-5]	[25-30]
Lindau	[5-10]	[5-10]	**[10-15]**	[30-35]	[15-20]	**[60-70]**	[0-5]	[5-10]	[5-10]	[0-5]	[15-20]
Ottokraftstoff 2020											
20 Minuten Fahrzeit (Anteile in %)											
Kressbronn	[5-10]	[5-10]	**[10-15]**	[25-30]	[10-15]	**[50-60]**	[0-5]	10,88	[5-10]	[0-5]	[20-25]
Lindau	[10-15]	[10-15]	**[20-25]**	[30-35]	[20-25]	**[80-90]**	[0-5]	[5-10]	[0-5]	[10-15]	[0-5]
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Left: 30-minute CA of OMV Kressbronn and Lindau. Right: 2020 shares, 20 and 30 minutes. Source: BKartA, B8-77/21.

- Three largest firms hold 50–90% with both 20- and 30-minute catchment areas

Case study: MOL/OMV Slovenija (cleared with remedies)

- EC case M.10438: notified May 2022, Phase II, cleared with conditions on 17 May 2023*
- MOL (No. 3, 53 stations) acquires OMV Slovenija (No. 2, 119); incumbent Petrol has 313
- Merged firm and Petrol together: more than 90% of the market
- Unilateral and coordinated effects (station prices are published in real time)
- National market, as prices are set nationally; but drivers refuel near home or work, so local effects were assessed with catchment areas

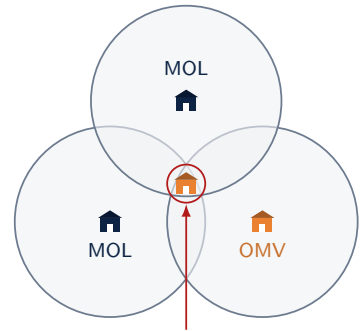
*Source: European Commission, M.10438 MOL/OMV Slovenija, decision of 17 May 2023; Competition Merger Brief 3/2023.

MOL/OMV: local assessment

- Supplier-centric catchment areas: 15-minute driving-time isochrones around each party station, the distance drivers are prepared to drive to refuel
- Presence-based local shares: all rival stations inside the isochrone, each weighted by its brand's average volume per station
- Screen: joint share above 40% and increment of more than 5 pp
  - **85** problematic areas out of **147** around the parties' regular stations
  - Often 3 to 2 or 4 to 3 retailers; small rivals mostly at the edge of the area
- The parties' "local upward pricing pressure" analysis (diversion ratios and margins) was not accepted because of data and method problems

MOL/OMV: remedy

- Divestment of **39** stations: 20 MOL and 19 OMV stations
- Fix-it-first: buyer Shell Adria named before clearance
- One divested station lowers the parties' share in every catchment area it lies in: 39 stations cure all 85 areas
- In each problematic area the merged firm still faces at least two rivals (Petrol and Shell)
- Shell becomes the new national No. 3, which also addresses coordinated effects



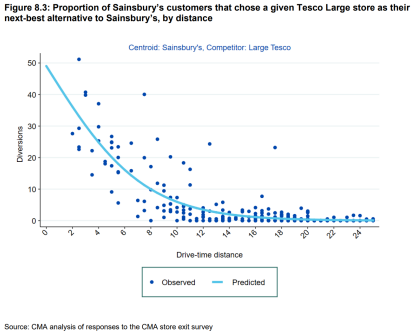
Case Study: Sainsbury's–Asda (blocked)

- Case before the UK Competition and Markets Authority (CMA)
- Top-5 UK grocery retailers: Tesco (£53.2bn), Sainsbury's (£29.0bn), Asda (£22.8bn), Amazon (£19.4bn), Morrisons (£17.6bn)
- **537** critical stores: the CMA found that prohibition was the only remedy
- A store exit survey allowed a more detailed analysis of CAs
  - CAs weighted by product differentiation, size and distance
  - GUPPIs by store (diversion)
  - Out-of-market constraints: more distant stores, online grocery delivery, non-supermarket retailers

Sainsbury's–Asda: store exit survey

- 20,000+ customers at 100 Sainsbury's and Asda stores
- Key question: where would you have shopped if this store had been closed?
- Also: would you still shop here if prices were 5% higher? (20–30% would not)
- Answers give diversion ratios between stores, used for weights and GUPPIs

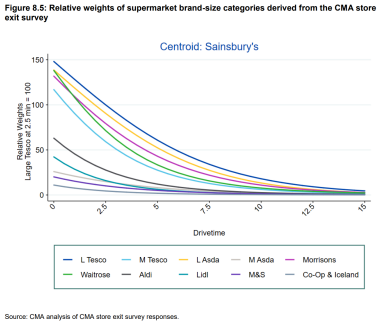
Sainsbury's–Asda

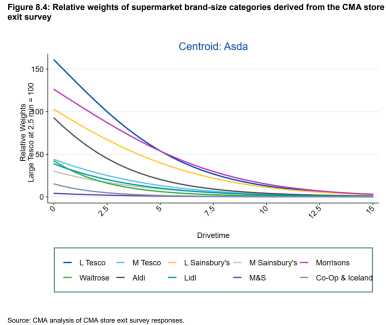


Share of Sainsbury's customers choosing a given large Tesco as next-best alternative, by drive time. Source: CMA (2019), Final Report, based on the CMA store exit survey.

- Little substitution to Tesco after 10–15 minutes of driving time

Sainsbury's–Asda





Relative weights of supermarket brand-size categories (large Tesco at 2.5 minutes = 100) for Sainsbury's (left) and Asda (right) centroids. Source: CMA (2019), Final Report.

- Rivals weighted by brand, size and driving time, not counted equally

Current case: EDEKA/tegut (cleared with conditions)

- Bundeskartellamt, 28 Sep 2026: EDEKA may take 178 of 202 tegut stores*
- Local markets around each store's catchment area: competition significantly impeded in 34 regional markets, 9 of them through dominance
- Remedy without a divestiture: 24 tegut stores are excluded from the deal, and the whole EDEKA group may not acquire or lease them for four years
  - These 24 stores fix all 34 markets: 24 directly, 10 through overlapping catchment areas; who runs them is left open
- No collective dominance of the big four (together more than 90% of sales)
- Still open: REWE's bid for further tegut stores (deadline 29 October 2026)
- *For discussion: does a four-year acquisition ban protect competition as well as a sale to a named buyer, as in MOL/OMV?*

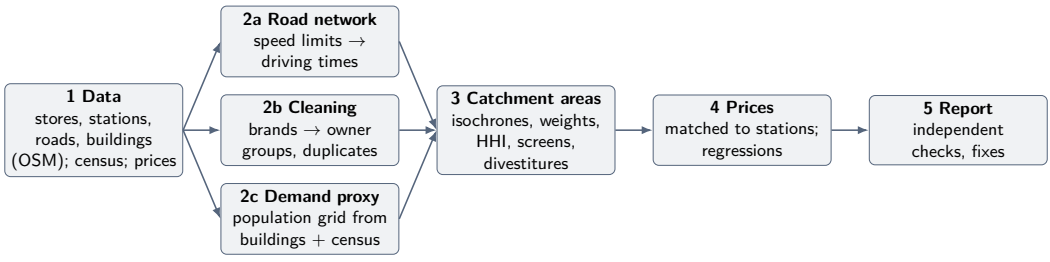
*Source: Bundeskartellamt, press releases of 31 July and 28 September 2026.

Examples of academic studies

- Catchment areas
  - Verboven, F. and B. Yontcheva (2024). Private monopoly and restricted entry: evidence from the notary profession. *Journal of Political Economy* 132(11), 3658–3707.
  - Fischer, K., S. Martin and P. Schmidt-Dengler (2026). The heterogeneous effects of entry on prices. *Journal of the European Economic Association* 24(5), 1626–1670.
- Clustering
  - Zheng, F. (2016). Spatial competition and preemptive entry in the discount retail industry. Working paper, Columbia Business School.

Demo with Agentic AI

A local market analysis in practice: the Brno demo



Brno demo: 213 supermarkets and 526 petrol stations from OpenStreetMap; road network with 93,667 nodes; population grid from 317,534 buildings and the 2021 census; crowd-sourced prices for 81 stations.

Complications

- Data access is the bottleneck: public map servers time out, a bulk download of the whole country took minutes instead of hours
- Judgement calls need checking: who owns which brand, which stores count, duplicates
- Method choices drive results: radius or driving time, weights, thresholds; hence robustness checks
- Missing quantities need proxies: buildings and the census for where people live; crowd-sourced prices are patchy and can be stale
- Independent checks before the numbers are trusted: recompute shares by hand, verify totals, read the maps

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